Transient Phenomena course (MPA-PRJ)

Assignment 1

Enter your BUT ID here, each digit in a separate cell.

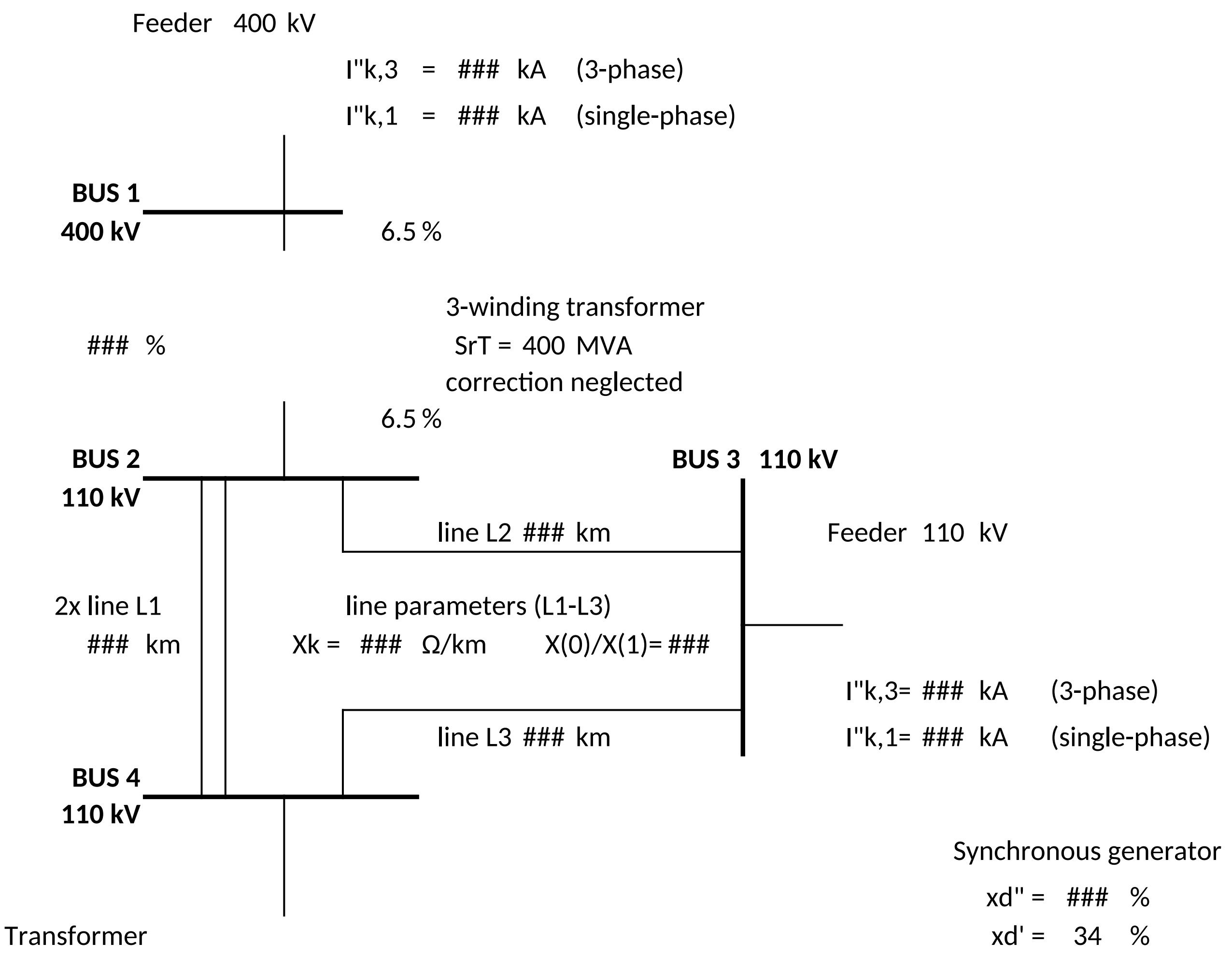
ID:						
	2	8	5	2	6	9

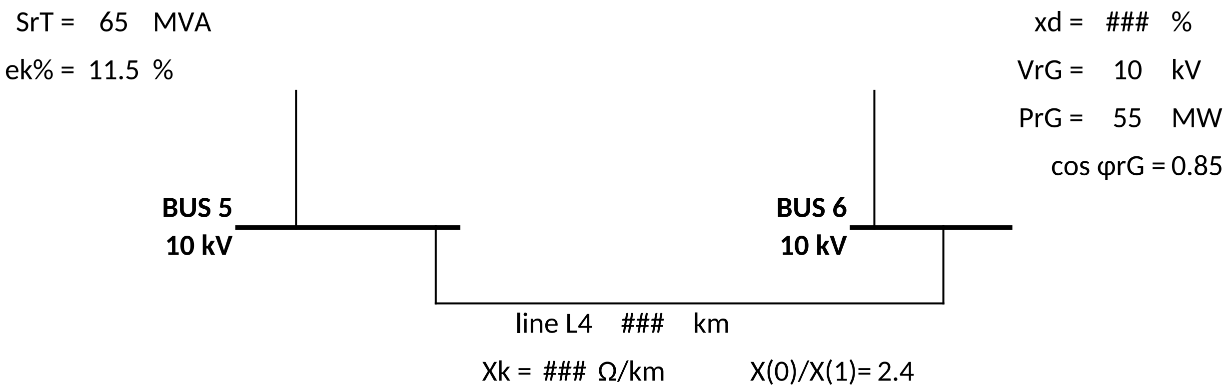
The individual assignment is then generated in the sheet "Assignment1".

Instructions:
The assignment should be either hand-written or done on computer, but you cannot combine these two! If combined, the grade is automatically zero.
If the assignment is hand-written, use only the blank sheets (no lines) and your writing has to be legible.
Always put formula first, then substitute the numerical values and give the result with corresponding unit.
You must draw equivalent impedance diagrams and indicate their simplifacation to the resulting impedance.
Calculation is done according to the current version of IEC 60909-0 standard. Correction factors are considered except for three-winding transformer.
Calculation is done in per unit system with a given base.
You must give the conclusion to the assignment.
The assignment solution has to be compiled as one pdf file and submitted via the elearning.
Submission deadline is mandatory and cannot be extended.
Teachers will answer only specific questions, they will not provide a step-by-step guide!
Teachers will not check your assignment before its submission.

ASSIGNMENT 1

Calculate the initial short circuit currents in kA in bus number **###** in the given network. Use only reactances. Correction factors are considered, except for 3-winding transformer. Voltage factor c = 1.1.
Calculation should be performed in per unit values with a base power o ### MVA.




SrT = 65 MVA
ek% = 11.5 %
xd = ### %
VrG = 10 kV
PrG = 55 MW
cos φrG = 0.85
BUS 5
10 kV
BUS 6
10 kV
line L4 ### km
Xk = ### Ω/km
X(0)/X(1)= 2.4

Assignment 1 – Solution

Bus 3 (110 kV) | Sbase = 100 MVA | c = 1.1

Element	Parameter	Value	Unit	Note
Task	Bus number to calculate	**3**		
Base	Base power Sbase	**100**	MVA	
Base	Voltage base (Bus 1)	**400**	kV	
Base	Voltage base (Bus 2,3,4)	**110**	kV	
Base	Voltage base (Bus 5,6)	**10**	kV	
Base	Zbase at 400 kV	**1600**	Ω	
Base	Zbase at 110 kV	**121**	Ω	
Base	Zbase at 10 kV	**1.0**	Ω	
Feeder Q1	Voltage	**400**	kV	
Feeder Q1	I"k,3	**18**	kA	
Feeder Q1	I"k,1	**14.5**	kA	
3WT	Rated power SrT	**400**	MVA	

3WT	ukHM (H-winding to M-winding)	**12.0**	%
3WT	ukHT (H-winding to T-winding)	**6.5**	%
3WT	ukMT (M-winding to T-winding)	**6.5**	%
2×L1	Length (each line)	**98**	km
L1–L3	Positive-seq reactance Xk	**0.31**	Ω/km
L1–L3	Zero-seq ratio X0/X1	**3.3**	—
L2	Length	**70**	km
L3	Length	**117**	km
Feeder Q2	Voltage (at Bus 3)	**110**	kV
Feeder Q2	I"k,3	**11.8**	kA
Feeder Q2	I"k,1	**7.9**	kA
Transformer T2	Rated power SrT	**65**	MVA
Transformer T2	ek%	**11.5**	%
Generator G	x"d (subtransient)	**15.0**	%
Generator G	x'd (transient)	**34**	%
Generator G	xd (synchronous)	**129**	%
Generator G	VrG (rated voltage)	**10**	kV

Generator G	PrG (rated power)	**55**	MW
Generator G	cos φrG	**0.85**	—
Line L4	Length	**14**	km
Line L4	Xk	**0.26**	Ω/km
Line L4	X0/X1	**2.4**	—

2. PER-UNIT CONVERSIONS OF NETWORK ELEMENTS

Element	Formula	Result (pu)	Description
Feeder Q1 (400 kV)			
Z1_Q1	c·Vn/(√3·I"k,3) = 1.1×400/(√3×18)	**14.113 Ω** → **0.00882 pu**	pos. seq.
Z0_Q1	√3·c·Vn/I"k,1 − 2·Z1 = √3×1.1×400/14.5 − 2×14.113	**24.333 Ω** → **0.01521 pu**	zero seq.
3-Winding Transformer (correction neglected)			
ukH	(ukHM+ukHT−ukMT)/2 = (12+6.5−6.5)/2	**6.0%**	ukH

ukM	(ukHM+ukMT−ukHT)/2 = (12+6.5−6.5)/2	**6.0%**	ukM
ukT	(ukHT+ukMT−ukHM)/2 = (6.5+6.5−12)/2	**0.5%**	ukT
xH	ukH% × Sbase/SrT = 0.06 × 100/400	**0.01500 pu**	xH [pu]
xM	ukM% × Sbase/SrT = 0.06 × 100/400	**0.01500 pu**	xM [pu]
xT	ukT% × Sbase/SrT = 0.005 × 100/400	**0.00125 pu**	xT [pu]

Lines (Xk = 0.31 Ω/km, X0/X1 = 3.3, Zbase = 121 Ω)

x1_2L1	0.31×98/(2×121)	**0.12554 pu**	pos. seq.
x0_2L1	3.3×0.31×98/(2×121)	**0.41427 pu**	zero seq.
x1_L2	0.31×70/121	**0.17934 pu**	pos. seq.
x0_L2	3.3×0.31×70/121	**0.59182 pu**	zero seq.
x1_L3	0.31×117/121	**0.29975 pu**	pos. seq.
x0_L3	3.3×0.31×117/121	**0.98918 pu**	zero seq.
x1_L2||L3	x1_L2·x1_L3/(x1_L2+x1_L3)	**0.11221 pu**	pos. seq.
x0_L2||L3	x0_L2·x0_L3/(x0_L2+x0_L3)	**0.37028 pu**	zero seq.

Feeder Q2 (110 kV)

Z1_Q2	c·Vn/(√3·I"k,3) = 1.1×110/(√3×11.8)	**5.920 Ω → 0.04893 pu**	pos. seq.

Z0_Q2	√3·c·Vn/I"k,1 − 2·Z1 = √3×1.1×110/7.9 − 2×5.920	**14.688 Ω** → **0.12139 pu**	zero seq.

Transformer T2 (YNd assumed)

xT2	ek%×Sbase/SrT = 0.115×100/65	**0.17692 pu**

Synchronous Generator

SrG	PrG/cosφrG = 55/0.85	**64.706 MVA**
sinφrG	√(1−0.85²)	**0.52678**

x"d (sys)	x"d%×Sbase/SrG = 0.15×100/64.706	**0.23182 pu**	system base
KG	c/(1+x"d×sinφrG) = 1.1/(1+0.15×0.52678)	**1.01945**	correction factor
x"dK	KG × x"d_sys = 1.01945×0.23182	**0.23633 pu**	corrected x"d
Line L4			
x1_L4	0.26×14/1.0	**3.6400 pu**	pos. seq.
x0_L4	2.4×0.26×14/1.0	**8.7360 pu**	zero seq.
Zgen_Bus4	x"dK + x1_L4 + xT2 = 0.23633+3.64+0.17692	**4.0532 pu**	gen. branch at Bus 4

ZT2_Bus4(0)	xT2 = 0.17692	**0.17692 pu**	zero seq. shunt

3. EQUIVALENT CIRCUIT REDUCTION — Node Admittance Method

Network topology: Q1→Bus1→[xH]→S→[xM]→Bus2→[2×L1]→Bus4→[L2‖L3]→Bus3 (fault bus)

Q2→Bus3 (direct) | Generator→L4→T2→Bus4 | S→[xT]→Bus3 (direct T-winding)

Star-point S replaces 3WT. Bus 1 folded as shunt at S: y_S = 1/(z1_Q1 + xH). T2 and generator folded as shunt at Bus 4.

Positive-Sequence Admittance Matrix Y1 (nodes: S, Bus2, Bus3, Bus4)

	S	**Bus 2**	**Bus 3**	**Bus 4**	**Diagonal composition**
S	908.647	-66.667	-800.000	0.000	1/(z1_Q1+xH) + 1/xM + 1/xT
Bus 2	-66.667	74.632	0.000	-7.966	1/xM + 1/x1_2L1

Bus 3	-800.000	0.000	829.350	-8.912	1/xT + 1/z1_Q2 + 1/x1_L2‖L3
Bus 4	0.000	-7.966	-8.912	17.125	1/x1_2L1 + 1/x1_L2‖L3 + 1/Zgen_Bus4

$Z_{1,Bus3} = [Y1^{-1}]_{33}$ = 0.016510 pu

Zero-Sequence Admittance Matrix Y0 (nodes: S, Bus2, Bus3, Bus4)

3WT grounding: YNynyn — T2: YNd (delta on LV side blocks zero-seq from generator)

	S	**Bus 2**	**Bus 3**	**Bus 4**	**Diagonal composition**
S	899.771	-66.667	-800.000	0.000	1/(z0_Q1+xH) + 1/xM + 1/xT
Bus 2	-66.667	69.081	0.000	-2.414	1/xM + 1/x0_2L1
Bus 3	-800.000	0.000	810.938	-2.701	1/xT + 1/z0_Q2 + 1/x0_L2‖L3

Bus 4	0.000	-2.414	-2.701	10.767	1/x0_2L1 + 1/x0_L2‖L3 + 1/xT2

$Z0,Bus3 = [Y0^{-1}]_{33}$ = 0.023479 pu

4. INITIAL SHORT-CIRCUIT CURRENT CALCULATION AT BUS 3 (110 kV)

Ibase = Sbase / (√3 · Vbase) = 100 / (√3 × 110) = 0.52486 kA

3-Phase Symmetrical Short-Circuit Current I"k,3

Formula: *I"k,3 = c · Ibase / Z1_Bus3*

Substitution: I"k,3 = 1.1 × 0.52486 / 0.016510

RESULT: **I"k,3 = 34.97 kA (3-phase)**

Single-Phase-to-Earth Short-Circuit Current I"k,1

Formula: *I"k,1 = √3 · c · Ibase / (2·Z1_Bus3 + Z0_Bus3)*

Substitution: I"k,1 = √3 × 1.1 × 0.52486 / (2×0.016510 + 0.023479)

= 1.0000 / (0.033020 + 0.023479) = 1.0000 / 0.056499

RESULT: **I"k,1 = 17.70 kA (single-phase-to-earth)**

5. SUMMARY

Quantity	Value	Unit	Comment
Z1_Bus3 (Thevenin positive-seq)	0.01651	pu	Obtained from $[Y1^{-1}]_{33}$
Z0_Bus3 (Thevenin zero-seq)	0.02348	pu	Obtained from $[Y0^{-1}]_{33}$
Ibase at 110 kV	0.52490	kA	Sbase/ (√3·Vbase)
I"k,3 (3-phase fault at Bus 3)	34.97000	kA	c·Ibase/Z1
I"k,1 (1-phase fault at Bus 3)	17.70000	kA	√3·c·Ibase/ (2Z1+Z0)

6. CONCLUSION

The initial short-circuit currents at Bus 3 (110 kV) are:

$I''_{k,3} = 34.97$ kA (3-phase)

$I''_{k,1} = 17.70$ kA (single-phase-to-earth)

The 3-phase fault current is mainly fed through the 3-winding transformer from the 400 kV feeder. The generator's contribution is small because of the high impedance of line L4 at the 10 kV level. The single-phase current is lower than the 3-phase value, which is consistent with $Z_0 > Z_1$.